
FunCQNet: A Functional Censored Quantile Neural Network for Predicting Long-Term Post-Transplant Kidney Survival

Jiaqi Men^{1*} Hua Liu^{2*} Yiming Tang^{3*} Jinhong You¹ Jianghu (James) Dong⁴ Jiguo Cao⁵

Abstract

Accurate survival prediction in kidney transplantation is critical yet challenging due to the complex interplay between functional biomarkers and patient characteristics under censoring. To address this, we propose a functional censored quantile neural network (FunCQNet), a novel framework that integrates deep neural networks with a censoring-adjusted sequential quantile loss to approximate interaction-dependent coefficient functions. We further introduce a conformal inference approach to rigorously assess the significance of functional-scalar interactions, ensuring interpretability alongside predictive power. Extensive simulations demonstrate that FunCQNet robustly recovers functional effects under varying noise and censoring levels. When applied to kidney transplant data, the model yields precise multi-quantile predictions and reveals clinically significant, age-dependent interaction patterns between donor type and recipient survival.

1. Introduction

Kidney transplantation is the preferred therapy for patients with end-stage renal disease (ESRD), substantially improving both survival and quality of life compared with dialysis (Wolfe et al., 1999; Tonelli et al., 2011). Each year, more than 80,000 kidney transplants are performed worldwide, making it the most common type of solid organ transplan-

tation (Lentine et al., 2022). Despite its success, kidney transplantation faces major challenges, including the persistent shortage of donor organs, risks of graft failure, and the need for lifelong immunosuppressive therapy (Hart et al., 2020; 2021; Lentine et al., 2023). Accurate projection of the post-transplant survival function is therefore essential for identifying high-risk recipients, guiding individualized postoperative management, improving long-term survival, and optimizing healthcare resource utilization.

Renal function progression, commonly assessed using estimated glomerular filtration rate (eGFR) records, is a key determinant of post-transplant prognosis (Keong et al., 2016). eGFR is a clinically validated marker of kidney health that integrates serum creatinine with age, sex, race, and body size (Hariharan et al., 2000). Prior studies have shown that post-transplant eGFR trajectories are strongly associated with long-term graft survival and patient mortality (Hariharan et al., 2000; Lentine et al., 2023). However, renal function is inherently dynamic, and single-timepoint measurements may fail to capture clinically relevant patterns. Clinical observations indicate that even patients with similar baseline characteristics may exhibit markedly different eGFR trajectories, which are closely linked to long-term survival. For example, patients whose eGFR declines rapidly in the first few years face a substantially higher risk of graft failure compared with those whose eGFR remains stable. This underscores the necessity of modeling eGFR as a functional clinical biomarker (Ramsay & Silverman, 2005; Wang et al., 2016; Chen et al., 2017; Kong et al., 2024). For example, as shown in Figure 1, the mean eGFR curve for patients who survived at least 16 years post-transplant is consistently higher than that of the other two recipient groups, who die or require re-transplantation within 6–10 or 10–16 years, consistent with the clinical interpretation that higher eGFR values indicate better renal function.

Transplantation survival outcomes are strongly influenced by recipient and donor demographics. Older age has been associated with reduced survival due to comorbidities and immunological challenges (Tonelli et al., 2011). Sex differences in immune response and metabolism may also influence graft outcomes (Hart et al., 2021). Matching between donor and recipient, including human leukocyte antigen

*Equal contribution ¹ School of Statistics and Data Science, Shanghai University of Finance and Economics, Shanghai, China ² School of Economics and Finance, Xi'an Jiaotong University, Xi'an, Shaanxi, China ³ School of Statistics and Mathematics, Shanghai Lixin University of Accounting and Finance, Shanghai, China ⁴ Department of Biostatistics, College of Public Health, University of Nebraska Medical Center, Omaha, Nebraska, USA ⁵ Department of Statistics and Actuarial Science, Simon Fraser University, Burnaby, BC, Canada. Correspondence to: Jiguo Cao <jiguo-ca@sfu.ca>.

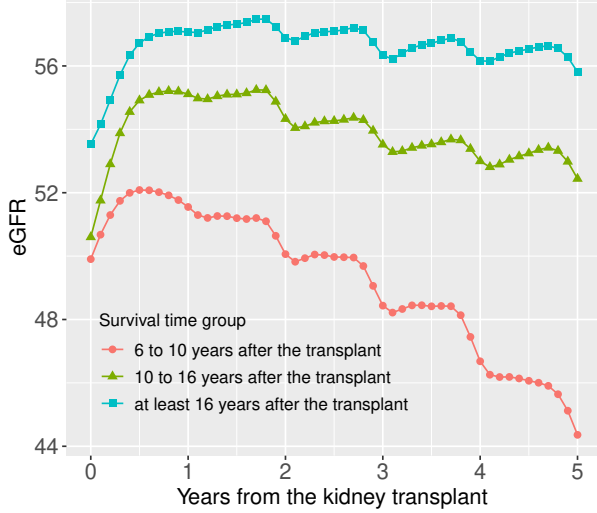


Figure 1. Mean eGFR trajectories for three groups of kidney transplant recipients: those who died or required re-transplantation within 6–10 years after transplant, those who died or required re-transplantation within 10–16 years, and those who survived at least 16 years post-transplant.

(HLA) compatibility, further impacts graft longevity and the likelihood of needing re-transplantation (Shi et al., 2018; Lentine et al., 2022; 2025). Moreover, compared with deceased donors, living-donor kidney transplantation typically yields better graft and patient survival outcomes, including higher 5-year survival rates and longer median graft survival.

For the analyses of time-to-event data with censored and functional covariates, the most commonly used models are the functional Cox proportional hazards model (Chen et al., 2010; Li & Ma, 2013; Gellar et al., 2015; Hao et al., 2021) and the functional accelerated failure time model (Liu et al., 2025). Some of these models simultaneously account for the effects of both functional and scalar covariates on the survival response. For example, Kong et al. (2018) employed a functional linear Cox regression model to investigate the association between in-hospital mortality and a combination of hippocampal radial distance functions and scalar demographic predictors in Alzheimer’s disease research, whereas Liu et al. (2025) applied a functional accelerated failure time model to examine the relationship between high-resolution physical activity trajectories, time-to-death outcomes, and several scalar covariates. Although these models provide a powerful framework for linking functional covariates with survival outcomes, they essentially perform a mean-type regression on the log-hazard scale. In clinical applications, researchers and clinicians are often interested not only in the average risk pattern but also in the heterogeneous effects across outcome quantiles, as these can reveal high-risk subgroups and inform more personalized treatment strategies.

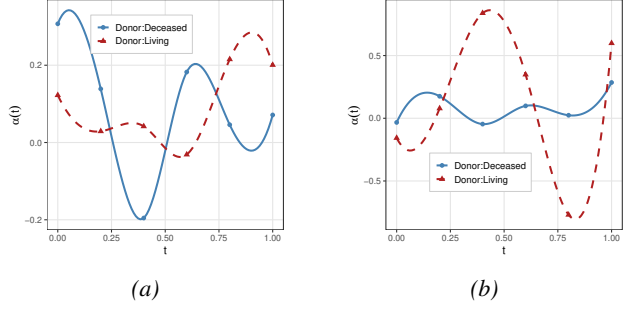


Figure 2. The estimated functional coefficient $\hat{\alpha}_\tau(t)$ from the functional censored quantile regression model without interactions (Jiang et al., 2020). The left panel displays $\hat{\alpha}_\tau(t)$ for young male recipients with age ≤ 18 when the quantile $\tau = 0.7$. The right panel displays $\hat{\alpha}_\tau(t)$ for young female recipients with age ≤ 18 when the quantile $\tau = 0.7$.

To address this, Jiang et al. (2020) considers a functional censored quantile regression framework: $Q_T(\tau|X, \mathbf{Z}; \alpha_\tau) = \exp \left\{ \mathbf{Z}^\top \beta_\tau + \int_{\mathcal{T}} X^\top(t) \alpha_\tau(t) dt \right\}$, which characterizes the conditional quantiles of survival time rather than its mean or hazard expectation, thereby providing a more comprehensive understanding of patient heterogeneity and tail risk behavior. While Jiang et al. (2020) simultaneously accounts for the effects of both functional and scalar covariates on survival time, similar to the aforementioned functional survival models that include scalar covariates, their framework also assumes an additive structure between functional and scalar effects, without accommodating potential interactions between them.

In our study, we focus on whether the effect of the functional clinical biomarker (eGFR) on survival time varies across levels of the scalar covariates $\mathbf{Z}$, motivating the need for a flexible modeling framework that can capture these interaction-dependent effects across the survival distribution. As a preliminary analysis, we stratify patients by donor type (living vs. deceased) and by sex (male vs. female). Separate functional censored quantile regression models are then fitted for each subgroup, and the resulting estimated functional effects are displayed in Figure 2. The results show that the estimated functional coefficients $\hat{\alpha}_\tau(t)$ differ significantly across the two groups with different donor types, a finding that will be further validated and supported in our subsequent inference analysis. This preliminary evidence suggests that it is crucial to investigate the interaction effect between the eGFR curve and the demographic and clinical covariates.

Therefore, to accurately identify high-risk patients and enable individualized post-transplant interventions, we integrate functional clinical biomarkers (eGFR trajectories) and key scalar covariates (age, sex, donor type), while accounting for their interactions. Furthermore, to capture dynamic heterogeneous interaction effects, which characterize how

the effect of eGFR on survival varies across outcome quantiles and patient characteristics, we propose a Functional Censored Quantile Neural Network (FunCQNet) model:

$$Q_T(\tau|\mathbf{Z}, \mathbf{X}; \alpha_\tau) = \exp \left\{ \beta_\tau + \int_{\mathcal{T}} \mathbf{X}^\top(t) \alpha_\tau(t, \mathbf{Z}) dt \right\}, \quad (1)$$

where T denotes post-transplant survival time, and $\mathbf{X}(t)$ is a p -dimensional vector of functional covariates defined on the domain $\mathcal{T}$, representing the post-transplant eGFR trajectory in this application. The q -dimensional vector of scalar covariates, $\mathbf{Z} = (Z_1, \dots, Z_q)^\top$, includes demographic characteristics and donor-related factors. And $Q_T(\tau|\mathbf{Z}, \mathbf{X}; \alpha_\tau) = \inf\{t_0 : \Pr(T \leq t_0|\mathbf{Z}, \mathbf{X}) \geq \tau\}$ denotes the τ th conditional quantile of T given $\mathbf{Z}$ and $\mathbf{X}$. The coefficient function $\alpha_\tau(t, \mathbf{Z}) = (\alpha_{\tau,1}(t, \mathbf{Z}), \dots, \alpha_{\tau,p}(t, \mathbf{Z}))^\top$ represents the τ -th quantile-specific effect function capturing how the influence of functional covariates varies over its domain $\mathcal{T}$ and scalar covariates $\mathbf{Z}$.

For models with functional covariates, the estimation and inference of the functional coefficients are crucial and of particular interest to us. Such coefficients are difficult to estimate and make inference, as they are infinite-dimensional functions that vary continuously over the functional domain. In conventional approaches, the functional coefficients are typically represented through basis expansions such as the spline method (de Boor, 2001; Cardot et al., 2003), Functional Principal Component Analysis (FPCA) (Yao et al., 2005; Hall et al., 2006; Dong et al., 2018), or alternatively mapped into a Reproducing Kernel Hilbert Space (RKHS) for modeling (Yuan & Cai, 2010). However, all these methods share a common limitation: they depend heavily on manual basis selection or truncation dimension choices. Moreover, in our setting, the interaction with scalar covariates leads to a multi-dimensional functional coefficient estimation problem, which substantially increases the modeling and computational complexity. As a result, traditional parametric and semi-parametric approaches are prone to the curse of dimensionality and are often ill-suited for estimating such multi-dimensional functional coefficients.

Recently, deep neural network (DNN)-based approaches have shown great promise in automatically learning flexible representations of complex functional relationships and alleviating the need for handcrafted basis functions (Zhang et al., 2023; Zhou et al., 2025). However, their applications to infinite-dimensional functional data remain relatively limited. Existing studies have primarily focused on how to incorporate functional data into neural networks, for instance, by treating discretized functions as high-dimensional vector inputs (Rossi et al., 2005) or by representing them through basis expansions (Yao et al., 2021; Wu et al., 2024). However, less attention has been paid to leveraging the expressive power of DNNs to address the estimation of multi-

dimensional functional coefficients. To tackle the curse of dimensionality induced by the complex interactions among functional and scalar covariates, here we use a deep neural network to estimate α to address the ‘‘curse of dimensionality’’. This formulation allows us to capture heterogeneous, potentially nonlinear associations between covariates and survival across outcome quantiles, offering a flexible and clinically relevant framework for risk stratification of patient mortality and other time-to-event outcomes.

This article has four major contributions. First, we propose a functional censored quantile regression model with interaction effects (FunCQNet), as specified in model (1), which flexibly characterizes the complex interdependence among functional covariates, scalar covariates, and a scalar response. Second, we employ deep neural networks to estimate the multi-dimensional nonlinear functional coefficients $\alpha_\tau(t, \mathbf{Z})$, which effectively alleviates the errors and challenges associated with basis truncation, tuning parameter selection, and the curse of dimensionality. Third, we integrate our FunCQNet with conformal inference, which is model-free, making it both simpler and more robust in practice (Lei et al., 2018; Jin & Candès, 2023; Liang et al., 2023; Bai et al., 2025). Last but not the least, we apply the proposed FunCQNet model to the five-year postoperative survival data of kidney transplant patients, providing precise survival predictions for patients with various clinical biomarkers and demographic characteristics. By capturing the heterogeneous functional effects and interactions, the proposed model enables a more nuanced understanding of post-transplant renal function trajectories and survival outcomes. It offers clinicians and researchers actionable insights for treatment optimization and postoperative care management.

The remainder of this paper is organized as follows. Section 2 describes our estimation and inference procedure. Section D illustrates some simulation studies that demonstrate the effectiveness of our approach and compare the empirical performance between our model and the conventional model. The proposed methods are then applied to the motivating dataset from the United Network for Organ Sharing/Organ Procurement Transplantation Network (UNOS/OPTN) in Section 3. In Section 4, we give some concluding remarks. All additional results of the simulation study can be found in the supplementary document. Additionally, we provide the codes needed to replicate the simulation studies at <https://github.com/Yiming-Tang/FunCQNet>.

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2. Estimation and Inference for FunCQNet

In this section, we first construct the martingale process underlying the FunCQNet model and, leveraging its zero-mean property, derive the corresponding estimating equations and objective function. We then represent the functional coefficient $\alpha_\tau(t, \mathbf{Z})$ using a neural network architecture tailored to capture complex time-covariate interactions, and describe the associated estimation algorithm. Finally, we develop a model-free and distribution-free inference procedure to assess the presence of interaction effects, enabling rigorous testing of whether interactions exist and identifying which scalar covariates contribute significantly to these effects.

2.1. Estimation

We begin by introducing the estimation strategy for the FunCQNet model (1). To estimate the functional coefficients in the FunCQNet model (1) with right-censored data, we begin by constructing an appropriate martingale process. Let C denote the censoring time, $Y = T \wedge C$ the observed time, and $\delta = I(T \leq C)$ the censoring indicator, where $\wedge$ represents the minimum operator and $I(\cdot)$ is the indicator function. We assume that the censoring variable C is conditionally independent of the event time T , given the scalar and functional covariates. Define the conditional cumulative distribution function (CDF) of T as $F_T(u | \mathbf{Z}, \mathbf{X}) = \Pr(T \leq u | \mathbf{Z}, \mathbf{X})$, and the associated cumulative hazard function as $\Lambda_T(u | \mathbf{Z}, \mathbf{X}) = -\log\{1 - F_T(u | \mathbf{Z}, \mathbf{X})\}$. Let $N(u) = I(Y \leq u, \delta = 1)$ be the counting process of observed failures. Then the process $M(u) = N(u) - \Lambda_T(u | \mathbf{Z}, \mathbf{X})$ forms a martingale, implying that $\mathbb{E}\{M(u) | \mathbf{Z}, \mathbf{X}\} = 0$, which serves as the foundation for deriving the estimating equations and the resulting objective function.

Based on the population-level martingale representation, we now derive the sample version required for estimation. Suppose that the data sample consists of n independent and identically distributed (i.i.d.) observations of $(Y, \delta, \mathbf{Z}, \mathbf{X}(t), t \in \mathcal{T})$, denoted by $\{(Y_i, \delta_i, \mathbf{Z}_i, \mathbf{X}_i(t), t \in \mathcal{T}), i = 1, \dots, n\}$. For subject $i = 1, \dots, n$, let $N_i(u) = I(Y_i \leq u, \delta_i = 1)$, $M_i(u) = N_i(u) - \Lambda_T(u | \mathbf{Z}_i, \mathbf{X}_i)$. The martingale property implies $\mathbb{E}\{M_i(u) | \mathbf{Z}_i, \mathbf{X}_i\} = 0$. Evaluating this property at the conditional quantile $Q_T(\tau | \mathbf{Z}_i, \mathbf{X}_i; \alpha_\tau)$ yields

$$\mathbb{E}\{M_i [Q_T(\tau | \mathbf{Z}_i, \mathbf{X}_i; \alpha_\tau)]\} = 0, \text{ for } i = 1, \dots, n. \quad (2)$$

Replacing the expectation by the empirical average yields the sample estimating equations:

$$\sum_{i=1}^n \{N_i[Q_T(\tau | \mathbf{Z}_i, \mathbf{X}_i; \alpha_\tau)] - \Lambda_T\{[Q_T(\tau | \mathbf{Z}_i, \mathbf{X}_i; \alpha_\tau)] \wedge Y_i | \mathbf{Z}_i, \mathbf{X}_i\}\} = 0. \quad (3)$$

In (3), note that $\Lambda_T\{[Q_T(\tau | \mathbf{Z}_i, \mathbf{X}_i; \alpha_\tau)] \wedge Y_i | \mathbf{Z}_i, \mathbf{X}_i\} = \int_0^\tau I\{Y_i \geq Q_T(s | \mathbf{Z}_i, \mathbf{X}_i; \alpha_s)\} dH(s)$, where $H(s) = -\log(1 - s)$, $0 \leq s < 1$. Because this integral involves unknown conditional quantile functions for $\tau_l < \tau$, we approximate it in practice using a partition $0 = \tau_0 < \tau_1 < \dots < \tau_L < 1$:

$$\begin{aligned} \hat{\Lambda}_T\{[Q_T(\tau | \mathbf{Z}_i, \mathbf{X}_i; \alpha_\tau)] \wedge Y_i | \mathbf{Z}_i, \mathbf{X}_i\} = \\ \sum_{\tau_l < \tau} \{H(\tau_{l+1}) - H(\tau_l) \times I\{Y_i \geq Q_T(\tau_l | \mathbf{Z}_i, \mathbf{X}_i; \alpha_{\tau_l})\}\}, \end{aligned} \quad (4)$$

with $Q_T(\tau_0 | \mathbf{Z}_i, \mathbf{X}_i; \alpha_{\tau_0}) = 0$.

Combining (3) with (4) yields the approximated sample estimating equation

$$\sum_{i=1}^n \left\{ N_i[Q_T(\tau | \mathbf{Z}_i, \mathbf{X}_i; \alpha_\tau)] - \sum_{\tau_l < \tau} \{H(\tau_{l+1}) - H(\tau_l)\} \times I\{Y_i \geq Q_T(\tau_l | \mathbf{Z}_i, \mathbf{X}_i; \hat{\alpha}_{\tau_l})\} \right\} = 0,$$

which is equivalent to minimizing the corresponding objective function with respect to $\alpha_\tau(\cdot, \cdot)$:

$$\begin{aligned} L_\tau = - \sum_{i=1}^n \{ \log(Y_i) - \log(Q_T(\tau | \mathbf{Z}_i, \mathbf{X}_i; \alpha_\tau)) \} \times \\ \left\{ N_i[Q_T(\tau | \mathbf{Z}_i, \mathbf{X}_i; \alpha_\tau)] - \sum_{\tau_l < \tau} \{H(\tau_{l+1}) - H(\tau_l)\} \times \right. \\ \left. I\{Y_i \geq Q_T(\tau_l | \mathbf{Z}_i, \mathbf{X}_i; \hat{\alpha}_{\tau_l})\} \right\}, \end{aligned} \quad (5)$$

with the initial condition $Q_T(\tau_0 | \mathbf{Z}_i, \mathbf{X}_i; \hat{\alpha}_{\tau_0}) = 0$ for $i = 1, \dots, n$. By sequentially solving (5) over $0 < \tau < \tau_L$, we obtain the estimator $\hat{\alpha}_\tau(\cdot, \cdot) = (\hat{\alpha}_{\tau,1}(\cdot, \cdot), \dots, \hat{\alpha}_{\tau,p}(\cdot, \cdot))^T$, $0 < \tau < \tau_L$.

Since the coefficient function $\alpha_\tau(\cdot, \cdot)$ in (5) is a multi-dimensional function varying with both the time index t and the scalar covariates $\mathbf{Z}$, the resulting optimization problem is intrinsically complex. This structure can lead to the curse of dimensionality when applying traditional nonparametric methods, such as basis-expansion approaches. To overcome this challenge, we adopt a neural network-based approximation for $\alpha_\tau(\cdot, \cdot)$, which enables flexible learning and efficient computation in these complex functional spaces.

2.2. FunCQNet

In this section, we develop a neural representation for the coefficient function. For each given quantile level τ , we design a neural network with an input dimension of $q + 1$ and an output dimension of p , as illustrated in the left panel of Figure 3. In practice, the functional covariate $\mathbf{X}(t)$ may be observed at irregular and individual-specific time

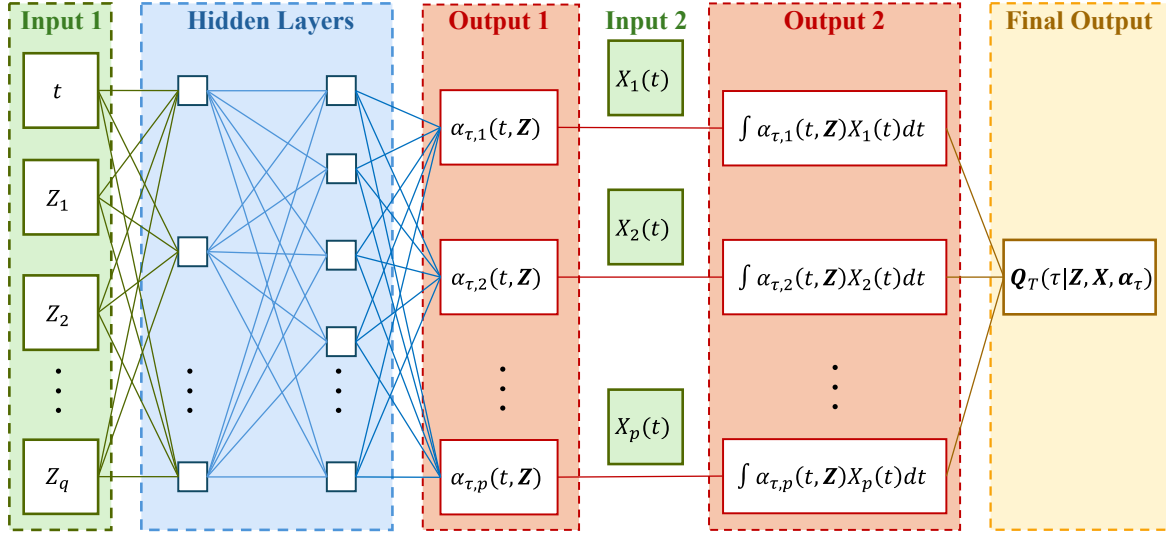


Figure 3. Overall architecture of the proposed FunCQNet model: $Q_T(\tau|\mathbf{Z}, \mathbf{X}; \alpha_\tau) = \exp\{\beta_\tau + \int_{\mathcal{T}} \mathbf{X}^\top(t) \alpha_\tau(t, \mathbf{Z}) dt\}$.

points, with varying numbers of sampling locations across subjects. To handle this heterogeneity, we explicitly include the time index t as part of the network input. This design ensures that the network output at *Output 1* corresponds one-to-one to the observed functional data $\mathbf{X}(t)$, thereby providing $\alpha_\tau(t, \mathbf{Z})$ evaluated at the same sampling points. By taking the element-wise product between $\alpha_\tau(t, \mathbf{Z})$ and $\mathbf{X}(t)$, and then performing integration over t and summation over p , we obtain the final network output $\hat{Q}_T(\tau|\mathbf{Z}, \mathbf{X}, \alpha_\tau)$. Importantly, we consider identifiability at the function level rather than at the level of neural-network parameters. Although different network weights may induce the same fitted function, the scientific target is the induced coefficient surface $\alpha_\tau(t, \mathbf{Z})$ and the corresponding conditional quantile function. In finite samples, the estimation is stabilized by validation-based tuning and early stopping.

Since the loss function L_τ for the τ -th quantile depends on the estimated $\hat{Q}_T(\tau_l|\mathbf{Z}, \mathbf{X}, \alpha_{\tau_l})$ values from all preceding levels $\tau_l < \tau$, we train an independent neural network for each quantile sequentially. During training, the estimated $\hat{Q}_T(\tau_l|\mathbf{Z}, \mathbf{X}, \alpha_{\tau_l})$ values from previous quantiles are concatenated to compute L_τ , enabling the model to leverage information from lower quantiles. The overall training procedure is summarized in Algorithm 1 (Appendix A) and the Adam optimizer (Kingma & Ba, 2015) is adopted. All model configurations and training hyper-parameters are tuned as follows. The number of neural network layers, the number of neurons in each layer, and training-related hyperparameters such as the number of epochs, batch size, learning rate, and early-stopping patience are all selected based on performance on a validation set split from the training data.

2.3. Conformal Inference for FunCQNet

Due to space limitations, the complete details of the conformal-inference-based testing procedure for interaction effects (including both the global interaction test and the covariate-specific tests) are provided in Appendix B.

3. Application on Kidney Transplant Data

In this paper, we apply the proposed FunCQNet framework to kidney transplant recipients to predict post-transplant survival. By leveraging patients' longitudinal biokarker, eGFR trajectories alongside key clinical characteristics, the model want ot capture heterogeneous graft function patterns and provides individualized, clinically meaningful survival estimates.

Among various postoperative indicators, the estimated glomerular filtration rate (eGFR), a continuous function measured over time, serves as a key marker for evaluating renal graft function and long-term recovery. Unlike static variables, eGFR captures the dynamic evolution of kidney function after transplantation. Meanwhile, demographic and clinical characteristics such as age, sex, and donor type are also known to influence survival outcomes. Importantly, their effects may interact with the functional trajectory of eGFR, implying that the prognostic impact of renal function can vary substantially across patient subgroups. Therefore, it is essential to model these interaction effects between functional and scalar covariates to enable individualized and more accurate survival predictions.

In this application, we employ data from the United Network for Organ Sharing/Organ Procurement and Transplantation Network (UNOS/OPTN) database, updated to

September 2020. The data consist of 49181 subjects with 34.75% censoring rate. We treat the post-transplant eGFR trajectories observed within the first five years as the functional clinical biomarker $X(t)$, with the domain of t defined as $\mathcal{T} = [0, 5]$. The scalar covariates include Age group $Z_1 \in \{< 18, [18, 40), [40, 60), \geq 60\}$, Sex $Z_2 \in \{\text{Male, Female}\}$, and Donor type $Z_3 \in \{\text{Living, Deceased}\}$. And the failure time T denotes the post-five-year survival time (in years) following kidney transplantation. To ensure data completeness and consistency, patients with multiple kidney transplants or those lost to follow-up within five years are excluded from the analysis. Specially, we consider the following FunCQNet framework:

$$Q_T(\tau | \text{Age, Sex, Donor, eGFR}; \alpha) = \exp \left\{ \beta + \int_{\mathcal{T}} \text{eGFR}(t) \alpha_{\tau}(t, \text{Age, Sex, Donor}) dt \right\}, \quad (6)$$

where β denotes the baseline log-survival quantile when $X(t)$ and Z effects are centered. $\alpha_{\tau}(\cdot, \dots, \cdot)$ is the functional coefficient that characterizes the time-varying and patient-specific effects.

As stated in Section 2.3, we split 80% of the whole dataset into the training subset and the remaining into the test subset. Then, we first test whether the functional coefficient $\alpha_{\tau}(t, \text{Age, Sex, Donor})$ in model (6) can be simplified to a functional censored quantile regression without interaction. Given that the p -values across the quantile levels $\tau = 0.3, 0.5, 0.7$ are ($< 1e-3$, $< 1e-3$, 0.019), the null hypothesis $H_0 : \alpha_{\tau}(t, \text{Age, Sex, Donor}) = \alpha_{\tau}(t)$ is rejected at the 5% significance level for all three quantiles. These results indicate that the functional coefficient varies not only over time t but also with some components of $\mathbf{Z}$, demonstrating that interaction effects are indeed present and that the proposed FunCQNet model provides a better fit for the kidney transplant data.

Table 1. p -value results for testing whether the multi-dimensional functional coefficient in Model (6) does not depend on the scalar covariates.

Scalar covariate	$\tau = 0.3$	$\tau = 0.5$	$\tau = 0.7$
Effect of Age	0.790	$< 1e-3$	0.003
Effect of Sex	0.600	$< 1e-3$	0.330
Effect of Donor types	$< 1e-3$	$< 1e-3$	0.006

Figures 4, 5 and 6 present the estimated functional coefficients for $\tau = 0.3, 0.5, 0.7$, respectively, based on the FunCQNet model (6). When $\tau = 0.3$, the functional coefficient differs only among pediatric patients (age < 18) with donor type, while no substantial differences are observed for any other patient subgroups. This pattern is also consistent with Table 1, where we examine which scalar covariates exhibit interaction with the eGFR. At the 5% significance level, the null hypotheses $H_{0,j} : \alpha_{\tau}(t, \mathbf{Z}) = \alpha_{\tau}(t, \mathbf{Z}_{(-j)})$ cannot be

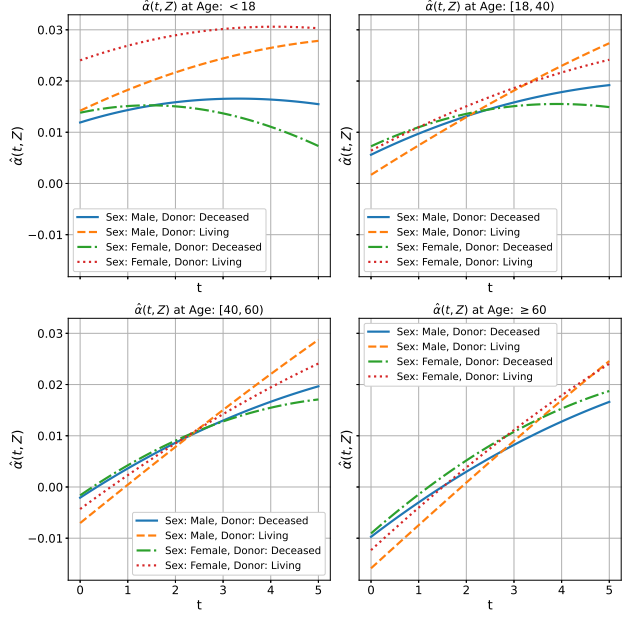


Figure 4. Estimated functional coefficient $\hat{\alpha}_{\tau}(t, \mathbf{Z})$ across sex and donor types for four age groups at quantile $\tau = 0.3$.

rejected for the covariates $Z_1 = \text{Age}$ and $Z_2 = \text{Sex}$, but are rejected for $Z_3 = \text{Donor Type}$ when $\tau = 0.3$. This finding suggests that, at high-risk survival levels (corresponding to the lower quantile of survival time), individual clinical characteristics no longer lead to meaningful differences in post-transplant outcomes. Clinically, patients with extremely poor prognosis often suffer from multiple, severe postoperative complications such as acute rejection, infection, delayed graft function, or metabolic failure that dominate the clinical course regardless of baseline characteristics. In such situations, the biological effect of donor type, age, or sex may be masked by the overwhelming severity of post-transplant complications (Kaplan & Meier-Kriesche, 2002).

As the survival risk decreases (moving to higher quantile levels), the effects of individual characteristics gradually become more prominent. For $\tau = 0.5$, the inference results presented in Table 1 reveal that all three scalar covariates significantly affect post-transplant survival at the 0.05 significance level. Specifically, the effect of donor type on survival time is substantially stronger in the younger group and gradually weakens with increasing age. Moreover, the differences between sexes are more pronounced within the younger age group. This phenomenon carries important implications for personalized medicine and clinical decision-making. It suggests that donor and sex characteristics play a dominant role in determining long-term graft survival for younger recipients, who may exhibit stronger immune responses and greater physiological sensitivity, whereas in older patients, other clinical factors may overshadow the

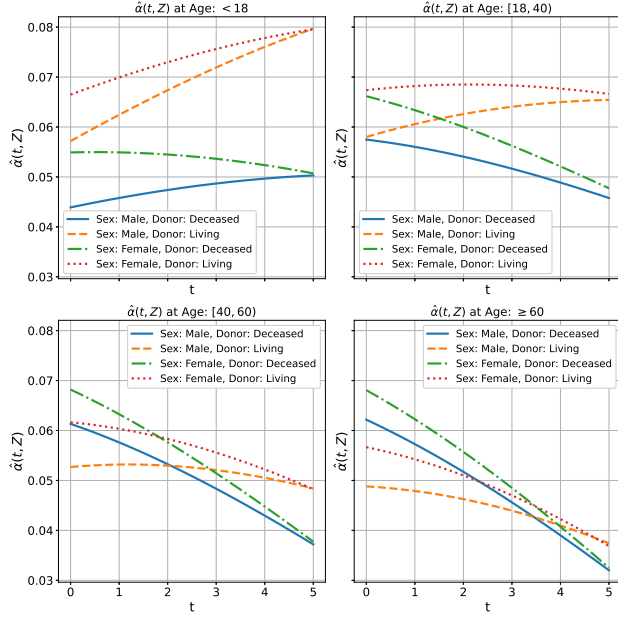


Figure 5. Estimated functional coefficient $\hat{\alpha}_\tau(t, \mathbf{Z})$ across sex and donor types for four age groups at quantile $\tau = 0.5$.

personal characteristic effect.

For $\tau = 0.7$, a similar pattern emerges: the inference test results ($p = 0.006 < 0.05$) indicate that the donor effect remains significant, suggesting that, regardless of the patient’s risk level, the type of donor consistently exerts a significantly non-negligible influence on postoperative survival time. The consistency of significant p -values (all < 0.05) across quantiles reinforces that tailoring transplantation decisions and postoperative management according to patient subgroups, particularly those defined by the significant covariates, could improve risk assessment, optimize donor-recipient matching, and ultimately enhance overall transplant success rates.

Despite these commonalities, distinct patterns arise when comparing $\tau = 0.3, 0.7$ and $\tau = 0.5$. The estimated functional coefficients display opposite temporal trends across the quantiles, indicating that the influence of eGFR on survival time varies along the survival distribution. For patients with relatively shorter and longer post-transplant survival ($\tau = 0.3$ and $\tau = 0.7$), the effect of eGFR on survival tends to increase over time, suggesting that sustained renal function plays an increasingly vital role in long-term prognosis. This pattern underscores the importance of incorporating survival-time heterogeneity through quantile regression. Unlike mean-based models that capture only the average trend, our quantile framework allows us to reveal how the influence of renal function differs between short-term and long-term survivors.

Taken together, these findings highlight the necessity of indi-

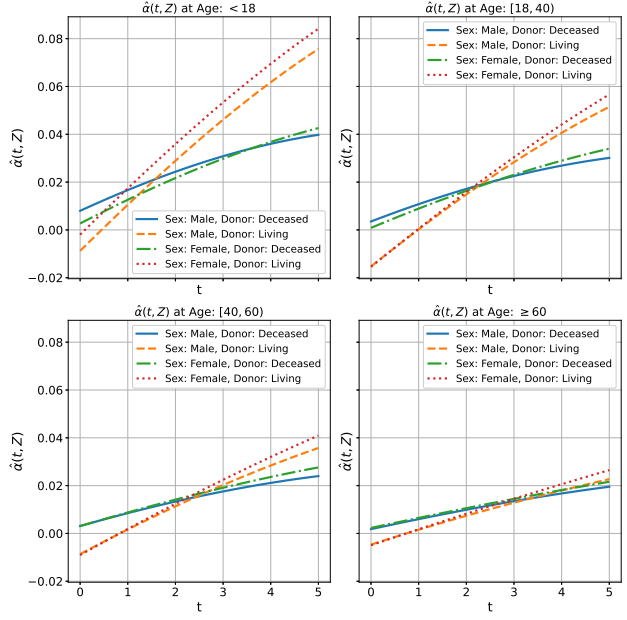


Figure 6. Estimated functional coefficient $\hat{\alpha}_\tau(t, \mathbf{Z})$ across sex and donor types for four age groups at quantile $\tau = 0.7$.

vidualized patient stratification and targeted post-transplant management. Recognizing such heterogeneous effects enables clinicians to better identify high-risk subgroups, personalize treatment intensity, and improve the precision of postoperative survival prediction. To validate the rationality of modeling interaction effects between functional and scalar covariates, we compare our FunCQNet model with (Jiang et al., 2020)’s model that treats scalar covariates purely as additive effects, without interaction. The results indicate that FunCQNet consistently outperforms the competing model across quantile levels, highlighting the benefit of explicitly modeling the interaction effects. Detailed quantitative comparisons are provided in Appendix C.

To further explore and demonstrate the importance of individualized patient stratification in clinical survival prediction, we consider four representative types of patients characterized by distinct eGFR trajectories. These four groups correspond to patients with (1) a stably high eGFR, (2) a rising eGFR, (3) a declining eGFR, and (4) a stably low eGFR. Their corresponding eGFR trajectories are illustrated in Figure 7.

By leveraging the FunCQNet, our framework provides simultaneous predictions across multiple quantile levels, rather than a single mean prediction. This allows us to construct individualized prediction intervals for survival time. In particular, the boxplots displayed in Figure 8(a)–8(d) are derived from the interquartile range (25%–75%) of the predicted survival times, offering a clear and interpretable summary of both central tendency and uncertainty

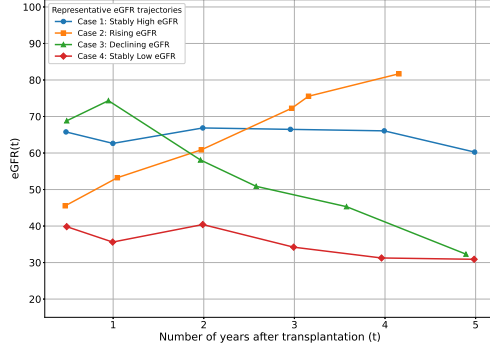
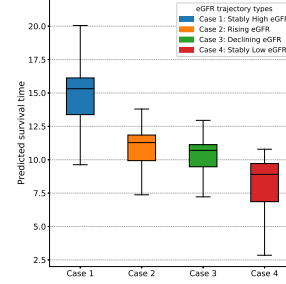


Figure 7. eGFR trajectories for four typical patient profiles.

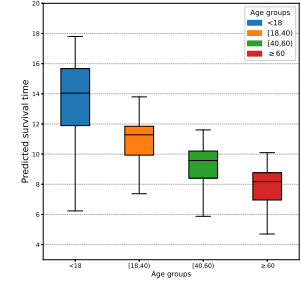
for different patient profiles.

Figure 8(a) presents boxplots of predicted survival times for four representative eGFR trajectories (shown in Figure 7) of a young male transplant recipient aged 18–40 with a deceased donor. This figure provides an intuitive visualization of the predicted survival intervals for each patient profile. Among the four representative patients, those with stably high postoperative eGFR levels exhibit the longest median predicted survival times. Notably, the corresponding boxplot is also wider, indicating greater variability in the estimated survival times and reflecting a broader prediction interval. This pattern suggests that while patients with stably high eGFR trajectories generally experience longer survival, individual outcomes may vary considerably. Conversely, patients with rising eGFR trajectories also tend to have relatively long survival times, but their prediction intervals are narrower, indicating more consistent and stable estimates. In contrast, patients with stably low eGFR levels exhibit shorter median predicted survival times, with wider boxplots, reflecting both the higher risk of adverse outcomes and increased uncertainty in survival predictions. Clinically, these observations align with the known association between early graft function and long-term post-transplant survival: patients who achieve moderate and stable kidney function postoperatively benefit from better overall prognosis, whereas impaired recovery or sustained low eGFR signals higher vulnerability and less predictable outcomes.

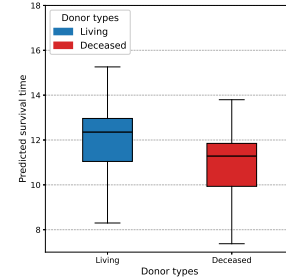
Figures 8(b)–8(d) display boxplots of the predicted survival times across age, sex, and donor-type subgroups, offering an intuitive visualization of both the central tendency and the variability of outcomes under rising eGFR trajectory group. As shown in Figure 8(b), predicted survival times exhibit a clear decreasing trend with increasing age. Younger patients (< 18) have the longest median predicted survival times, while older patients show progressively shorter survival. Figure 8(c) highlights a pronounced effect of donor type: recipients of living donor kidneys demonstrate longer median predicted survival compared to those receiving deceased donor kidneys, consistent with clinical evidence that living



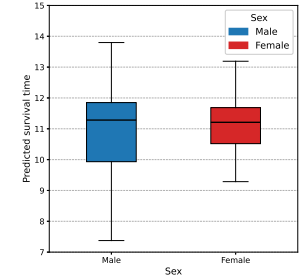
(a) Predicted survival times for a young male transplant recipient aged 18–40 with a deceased donor based on four representative eGFR trajectories (shown in Figure 7).



(b) Predicted survival times for male transplant recipients with a rising eGFR trajectory and a deceased donor across four age groups.



(c) Predicted survival times for a young male transplant recipient aged 18–40 with a rising eGFR trajectory and either a living or deceased donor.



(d) Predicted survival times for young male and female transplant recipients aged 18–40 with a rising eGFR trajectory and a deceased donor.

Figure 8. Boxplots of survival times for transplant recipients across different clinical biomarkers and demographic covariates.

donor transplants generally achieve better graft function and long-term outcomes. In contrast, Figure 8(d) shows that predicted survival times are similar between male and female patients, which is consistent with the previously estimated functional coefficients and the inference results indicating that sex is not a significant predictor of post-transplant survival. Interestingly, however, the predicted survival distribution for female patients is noticeably narrower than that of male patients. In other words, although the expected survival time does not differ substantially between sexes, the model provides more concentrated and precise predictions for females. From a clinical perspective, this pattern is plausible. Female transplant recipients generally demonstrate more stable immune responses and lower rates of certain post-transplant complications, as reported in previous studies (Katz-Greenberg & Shah, 2022), which may contribute to reduced heterogeneity in long-term survival outcomes. This observation suggests that while sex does not shift the overall survival level, female patients exhibit less variability in predicted prognosis, allowing clinicians to make more confident survival assessments in this subgroup.

4. Conclusions

In this study, we developed a functional censored quantile neural network (FunCQNet) to predict post-transplant survival times using longitudinal eGFR trajectories together with key clinical covariates, e.g., age, sex, and donor type. By jointly modeling censored survival outcomes and time-varying functional biomarkers within a deep quantile regression framework, FunCQNet enables individualized survival prediction with both survival estimates and associated uncertainty quantification across multiple quantile levels.

FunCQNet can flexibly capture heterogeneous patient trajectories and clinically meaningful interactions between functional and scalar covariates, while supporting interpretable inference on covariate contributions to survival risk. Empirically, the resulting predictions align well with established clinical knowledge: younger recipients and patients receiving living donor kidneys are associated with longer predicted survival, whereas age-related decline and deceased donor status correspond to higher risk. Although FunCQNet treats observed eGFR measurements as discretized functional trajectories and accommodates irregular sampling through time-indexed evaluation, substantial or structured measurement error could affect the estimated functional and interaction effects. Explicitly accounting for such error through calibration, smoothing, or latent trajectory modeling remains an important direction for future work. These results demonstrate the practical utility of FunCQNet for individualized risk stratification, personalized treatment planning, and informed clinical decision-making in kidney transplantation.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be

specifically highlighted here.

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A. Implementation Details

All experiments are implemented in PyTorch (Paszke et al., 2019) and conducted on a single NVIDIA A100 GPU with 80GB memory. The ADAM optimizer (Kingma & Ba, 2015) is used across all experiments. The longest running time among all experiments is less than 1.5 hours. The specific Algorithm can be found in Algorithm 1. The codes and hyper-parameters needed to replicate the simulation studies can be found at <https://github.com/Yiming-Tang/FunCQNet>.

Algorithm 1 Algorithm for training the proposed functional censored quantile neural network (FunCQNet) model (1): $Q_T(\tau|\mathbf{Z}, \mathbf{X}; \alpha_\tau) = \exp\{\beta_\tau + \int_T \mathbf{X}^\top(t) \alpha_\tau(t, \mathbf{Z}) dt\}$.

Require: $\mathbf{X}, \mathbf{Z}, Y, \delta, \{\tau_0, \tau_1, \dots, \tau_L\}$, Objective Function L_{τ_l} , $l = 1, \dots, L$, Number of Layers, Neurons per Layer, Epochs, Batch Size, Early Stop, Threshold for Minimum Improvement, Learning Rate, Dropout (Optional)

Ensure: The set of $\hat{\beta} = \{\hat{\beta}_{\tau_l}\}_{l=1}^L$ and parameters defining the neural networks $\hat{\alpha} = \{\hat{\alpha}_{\tau_l}(\cdot, \cdot)\}_{l=1}^L$

```

1:  $\hat{\mathbf{Q}}_T \leftarrow []$ 
2: for  $l = 1$  to  $L$  do
3:   Initialize  $\beta_{\tau_l}$  and define the DNN  $\alpha_{\tau_l}(\cdot, \cdot)$ 
4:   Train the DNN  $\hat{\alpha}_{\tau_l}(\cdot, \cdot)$  and update  $\hat{\beta}_{\tau_l}$  with  $L_{\tau_l}$  and  $\hat{\mathbf{Q}}_T$ 
5:   Output  $\hat{\mathbf{Q}}_T(\tau_l|\mathbf{Z}, \mathbf{X}, \hat{\alpha}_{\tau_l})$  and update  $\hat{\mathbf{Q}}_T \leftarrow [\hat{\mathbf{Q}}_T, \hat{\mathbf{Q}}_T(\tau_l|\mathbf{Z}, \mathbf{X}, \hat{\alpha}_{\tau_l})]$ 
6: end for
7: return  $\hat{\beta}$  and  $\hat{\alpha}$ 
    
```

B. Conformal Inference for FunCQNet

This section focuses on assessing whether the interaction effects, represented by the scalar covariates Z_j , $j = 1, \dots, q$, are statistically significant. In other words, the null hypothesis can be expressed as follows:

$$H_{0,j} : \alpha_\tau(t, \mathbf{Z}) = \alpha_\tau(t, \mathbf{Z}_{(-j)}), \text{ for any } t \text{ under a given } \tau,$$

where in each $\mathbf{Z}_{(-j)} = (Z_1, \dots, Z_{j-1}, Z_{j+1}, \dots, Z_p)^\top$, we remove the j th scalar covariate, $j = 1, \dots, q$.

We propose a testing procedure based on conformal inference (Lei et al., 2018). We also adopt a data-splitting strategy, splitting the whole dataset $\mathcal{I} = \{1, 2, \dots, n\}$ into the training subset $\mathcal{I}_1 \subseteq \mathcal{I}$ and test subset $\mathcal{I}_2 \subseteq \mathcal{I}$, where $\mathcal{I}_1 \cup \mathcal{I}_2 = \mathcal{I}$ and denote by $\mathcal{D}_k = \{(\mathbf{X}(\cdot), \mathbf{Z}_i, Y_i) : i \in \mathcal{I}_k, k = 1, 2\}$ the data samples in each index set. To test $H_{0,j}$, we fit on data $\{\mathbf{X}_i(\cdot), \mathbf{Z}_i, Y_i : i \in \mathcal{I}_1\}$ to get the estimator $\hat{\alpha}_\tau(\cdot, \cdot)$ and we refit on data $\{\mathbf{X}_i(\cdot), \mathbf{Z}_{i,(-j)}, Y_i : i \in \mathcal{I}_1\}$ to get the estimator $\hat{\alpha}_{\tau,(-j)}(\cdot, \cdot)$. For each sample $i \in \mathcal{I}_2$, denote the prediction loss of scalar covariate Z_j by $\Delta_j(\mathbf{X}_i, \mathbf{Z}_i, Y_i; \tau) = |\eta(\mathbf{X}_i, \mathbf{Z}_i, Y_i; \hat{\alpha}_\tau) - \eta(\mathbf{X}_i, \mathbf{Z}_{i,(-j)}, Y_i; \hat{\alpha}_{\tau,(-j)})|$,

where the loss value for $i \in \mathcal{I}_2$, evaluated using the estimators $\hat{\alpha}_\tau(\cdot, \cdot)$ and $\hat{\alpha}_{\tau,(-j)}(\cdot, \cdot)$ are denoted respectively by

$$\begin{aligned} \eta(\mathbf{X}_i, \mathbf{Z}_i, Y_i; \hat{\alpha}_\tau) &= \left\{ \sum_{\tau_l < \tau} I\{Y_i \geq Q_T(\tau_l|\mathbf{Z}_i, \mathbf{X}_i; \hat{\alpha}_{\tau_l})\} \times \{H(\tau_{l+1}) - H(\tau_l)\} \right. \\ &\quad \left. - N_i[Q_T(\tau|\mathbf{Z}_i, \mathbf{X}_i; \hat{\alpha}_\tau)] \right\} \times \{\log(Y_i) - \log(Q_T(\tau|\mathbf{Z}_i, \mathbf{X}_i; \hat{\alpha}_\tau))\}, \\ \eta(\mathbf{X}_i, \mathbf{Z}_{i,(-j)}, Y_i; \hat{\alpha}_{\tau,(-j)}) &= \left\{ \sum_{\tau_l < \tau} I\{Y_i \geq Q_T(\tau_l|\mathbf{Z}_i, \mathbf{X}_i; \hat{\alpha}_{\tau_l,(-j)})\} \times \{H(\tau_{l+1}) - H(\tau_l)\} \right. \\ &\quad \left. - N_i[Q_T(\tau|\mathbf{Z}_i, \mathbf{X}_i; \hat{\alpha}_{\tau,(-j)})] \right\} \times \{\log(Y_i) - \log(Q_T(\tau|\mathbf{Z}_i, \mathbf{X}_i; \hat{\alpha}_{\tau,(-j)}))\}. \end{aligned}$$

Thus, testing $H_{0,j}$ is equivalent to examining whether the mean of the random variable $\Delta_j(\mathbf{X}_i, \mathbf{Z}_i, Y_i; \tau)$ equals zero, i.e.,

$$H_{0,j} : \gamma_j(\tau) = \mathbb{E}(\Delta_j(\mathbf{X}_i, \mathbf{Z}_i, Y_i; \tau)|\mathcal{D}_1) \geq 0, \quad H_{1,j} : \gamma_j(\tau) < 0.$$

Following Lei et al. (2018), the rejection region for this test is given by

$$\{\sqrt{|\mathcal{I}_2|} \hat{\gamma}_j(\tau) / s_j(\tau) < -z_\alpha\},$$

where the sample mean $\hat{\gamma}_j(\tau) = 1/|\mathcal{I}_2| \sum_{i \in \mathcal{I}_2} \Delta_j(\mathbf{X}_i, \mathbf{Z}_i, Y_i; \tau)$, $s_j^2(\tau)$ is the sample variance of $\Delta_j(\mathbf{X}_i, \mathbf{Z}_i, Y_i; \tau)$, $i \in \mathcal{I}_2$ and $z_{\alpha/2}$ is the $1 - \alpha/2$ quantile of the standard normal distribution.

Specifically, we can also test whether the FunCQNet model (1) simplifies to a functional censored quantile regression model without interaction:

$$Q_T(\tau|\mathbf{X}; \alpha_\tau) = \exp \left\{ \beta_\tau + \int_{\tau} \mathbf{X}^\top(t) \alpha_\tau(t) dt \right\}. \quad (7)$$

That is, we test the null hypothesis

$$H_0 : \alpha_\tau(t, \mathbf{Z}) = \alpha_\tau(t), \text{ for any } t \text{ under a given } \tau,$$

by replacing $\{\mathbf{X}_i(\cdot), \mathbf{Z}_{i,(-j)}, Y_i\}, i \in \mathcal{I}_1$ and $\hat{\alpha}_{\tau,(-j)}(\cdot, \cdot)$ by $\{\mathbf{X}_i(\cdot), Y_i\}, i \in \mathcal{I}_1$ and $\hat{\alpha}_{\tau,(-\mathbf{Z})}(\cdot, \cdot)$, respectively. The other steps follow the same procedure as the test for $H_{0,j}$.

In summary, we should first test whether the FunCQNet model (1) can be reduced to the functional censored quantile regression model (7) without interaction under the null hypothesis H_0 . If H_0 is rejected, indicating that the interaction effects are significant, we then proceed to test the significance of each scalar covariate through the hypotheses $H_{0,j}$.

C. Additional Application Results

To further validate the necessity of modeling interaction effects between functional and scalar covariates, we conduct an additional comparison between the proposed FunCQNet model and the approach of Jiang et al. (2020). The latter treats scalar covariates purely as additive effects and does not account for interactions between scalar covariates and functional predictors.

Figure 9 reports the prediction losses of both methods across a range of quantile levels. Under all considered quantiles, FunCQNet consistently achieves lower prediction loss than the competing model. Notably, the performance gap is most pronounced around the median quantile ($\tau = 0.5$), where FunCQNet reduces the prediction loss by nearly 60% compared to the additive model.

These results provide strong empirical evidence that explicitly modeling functional–scalar interaction effects substantially improves predictive performance. In the context of kidney transplant survival analysis, this suggests that patient-specific scalar characteristics can meaningfully modulate the impact of longitudinal functional biomarkers, and that such heterogeneity cannot be adequately captured by purely additive models.

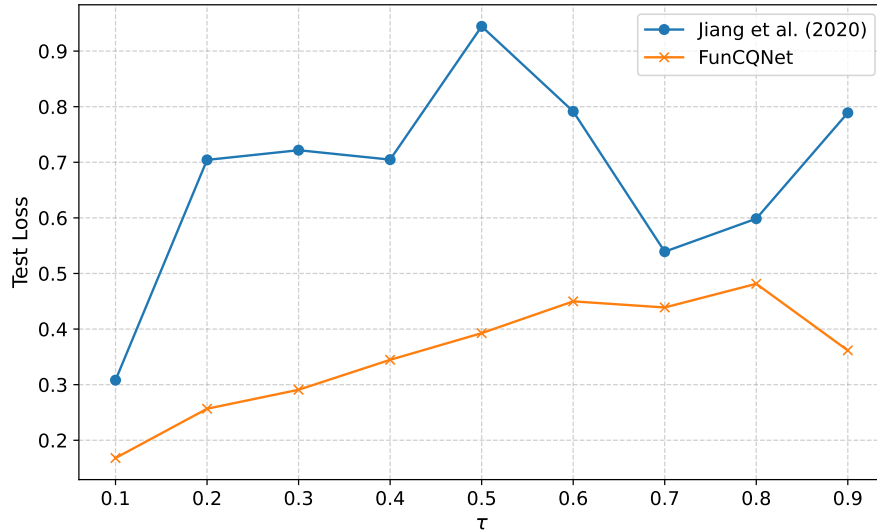


Figure 9. Comparison of prediction loss under various quantile levels between the proposed FunCQNet model and the additive functional model of Jiang et al. (2020).

In addition to the quantile prediction loss reported above, we further evaluate the survival prediction accuracy on the held-out

test set using the concordance index (C-index). Although FunCQNet is primarily designed to estimate conditional survival quantiles rather than a conventional survival risk score, the C-index provides a useful supplementary measure of survival-risk ranking. Specifically, we use the predicted median survival time to derive the ranking score.

As shown in Table 2, the full FunCQNet model achieves a C-index of 0.600. In comparison, the ablated model without the interaction term obtains a C-index of 0.574, while the versions without age, sex, and donor type obtain C-index values of 0.583, 0.589, and 0.580, respectively. These results suggest that the full model improves survival-ranking performance over the ablated variants, although the improvement is moderate. For further comparison, we also fit a standard Cox proportional hazards model on the same dataset, which yields a C-index of 0.580. Thus, FunCQNet is at least competitive with the Cox model from the perspective of survival-risk ranking.

We report the C-index as a supplementary metric because it only requires a ranking score and is therefore compatible with quantile-based survival prediction. By contrast, time-dependent AUC, calibration curves, and Brier scores typically require a well-defined survival probability function over time, which is not the primary output of our quantile-based framework. Moreover, the C-index itself has limitations: it mainly evaluates ranking performance and does not directly assess the accuracy of conditional quantile estimation or the heterogeneous functional effects that are the main targets of this paper. Therefore, we view these real-data results as supportive empirical evidence rather than as the sole basis for evaluating the proposed method.

Table 2. C-index comparison on the held-out test set.

Method	C-index
FunCQNet	0.600
FunCQNet without interaction	0.574
FunCQNet without age	0.583
FunCQNet without sex	0.589
FunCQNet without donor type	0.580
Cox model	0.580

D. Simulation Studies

In this section, we present two simulation studies to assess the finite sample performance of the proposed estimation and inference method. Additionally, we compare its performance with the model without interaction.

D.1. Simulation I

The first simulation study evaluates the finite sample performance of our estimators described in Section 2. We consider two functional predictors $\mathbf{X}_i(t) = (X_{i1}(t), X_{i2}(t))^T$ and vector predictors $\mathbf{Z}_i = (Z_{i1}, Z_{i2}, Z_{i3})$, where $X_{i1}(t)$ is constructed as $X_{i1}(t) = \sum_{k=1}^{\kappa} \zeta_{ik} B_k(t)$, $\kappa = 20$, $\zeta_{ik} \stackrel{iid}{\sim} N(0, \sigma_{\zeta}^2)$, $\sigma_{\zeta}^2 = 4$ and $B_k(t)$ is the cubic B-spline basis function defined on $\kappa - 4$ equally spaced interior knots in $[0, 1]$. And $X_{2i}(t)$ is given by $X_{2i}(t) = |\sum_{k=1}^{\kappa} \xi_k U_{ik} \phi_k(t)|$, $\phi_1(t) = 1$, $\phi_k(t) = \sqrt{2} \cos \{(k-1)\pi s\}$ for $k > 1$, with $\xi_k = (-1)^{k+1} k^{-1}$, $U_{ik} \stackrel{iid}{\sim} U[-3, 3]$, $Z_{i1} \sim U(1, 2)$ and $Z_{i2}, Z_{i3} \sim \text{Bernoulli}(0.5)$.

Here, we consider three scenarios to evaluate the performance of our models. In Scenario 1 and Scenario 2, the functional coefficient $\alpha_d(\cdot, \cdot)$, $d = 1, 2$ varies with $\mathbf{Z}$, indicating that the effect of X on $\log(T)$ is modulated by Z . In Scenario 1, we impose homoscedastic error terms, whereas in Scenario 2, we introduce heteroscedastic errors whose conditional variance depends on X . The latter design provides a clearer demonstration of the capability of quantile regression to capture asymmetry and heteroscedasticity in the conditional distribution, thereby offering a setting that more closely resembles the characteristics of complex real-world data. In contrast, Scenario 3 considers a relatively simple setting where the functional coefficient $\alpha_d(\cdot, \cdot)$ does not vary with Z . Scenario 3 is designed to assess whether our model can retain a reliable estimation performance under a simplified specification.

- **Scenario 1 (homogeneous, with interaction)** The true data is generated as

$$\log(T_i) = \beta + \int_0^1 X_{1i}(t) \psi_{11}(t, \mathbf{Z}) dt + \int_0^1 X_{2i}(t) \psi_{21}(t, \mathbf{Z}) dt + \epsilon_i - F_{\epsilon}^{-1}(\tau),$$

thus $\alpha_1(t, \mathbf{Z}) = \psi_{11}(t, \mathbf{Z})$, and $\alpha_2(t, \mathbf{Z}) = \psi_{21}(t, \mathbf{Z})$, where $\psi_{11}(t, \mathbf{Z}) = 2/3\{Z_{i1}t^3 + Z_{i2}\exp(-t) + Z_{i3}\cos(t)\}$, $\psi_{21}(t, \mathbf{Z}) = 1/5\{Z_{i1}t^2 + 2Z_{i2}\cos(3\pi t) + 3Z_{i3}\exp(-t)\}$, $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma_\epsilon^2)$, and $\sigma_\epsilon^2 = 0.01$, $\beta = 0.5$. F_ϵ is the cumulative distribution function of ϵ .

- **Scenario 2 (heterogeneous, with interaction)** The true data is generated as

$$\log(T_i) = \beta + \int_0^1 X_{1i}(t)\psi_{11}(t, \mathbf{Z})dt + \int_0^1 X_{2i}(t)\psi_{21}(t, \mathbf{Z})dt + \left\{ \int_0^1 X_{2i}(t)\psi_{22}(t, \mathbf{Z})dt \right\} \epsilon_i,$$

thus $\alpha_1(t, \mathbf{Z}) = \psi_{11}(t, \mathbf{Z})$, and $\alpha_{\tau,2}(t, \mathbf{Z}) = \psi_{21}(t, \mathbf{Z}) + \psi_{22}(t, \mathbf{Z})F_\epsilon^{-1}(\tau)$, where $\psi_{11}(t, \mathbf{Z}) = 2/3\{Z_{i1}t^3 + Z_{i2}\exp(-t) + Z_{i3}\cos(t)\}$, $\psi_{21}(t, \mathbf{Z}) = 1/5\{Z_{i1}t^2 + 2Z_{i2}\cos(3\pi t) + 3Z_{i3}\exp(-t)\}$, $\psi_{22}(t, \mathbf{Z}) = 1$, $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma_\epsilon^2)$, and $\sigma_\epsilon^2 = 0.01$, $\beta = 0.5$.

- **Scenario 3 (heterogeneous, without interaction)** The true data is generated as

$$\log(T_i) = \beta + \int_0^1 X_{1i}(t)\psi_{11}(t)dt + \int_0^1 X_{2i}(t)\psi_{21}(t)dt + \left\{ \int_0^1 X_{2i}(t)\psi_{22}(t)dt \right\} \epsilon_i,$$

thus $\alpha_1(t, \tau) = \psi_{11}(t)$, and $\alpha_{\tau,2}(t) = \psi_{21}(t) + \psi_{22}(t)F_\epsilon^{-1}(\tau)$, where $\psi_{11}(t) = 2/3\{t^3 + \exp(-t) + \cos(t)\}$, $\psi_{21}(t) = 1/5\{t^2 + 2\cos(3\pi t) + 3\exp(-t)\}$, $\psi_{22}(t) = 1$, $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma_\epsilon^2)$, and $\sigma_\epsilon^2 = 0.01$, $\beta = 0.5$.

In each simulation case, we perform $B = 100$ simulation runs with sample size $n = 50000$ and censoring rates (0%, 10%, 30%, 50%). We use 70% of the simulated data in each simulation run to compute our estimates under a particular hyperparameter configuration (e.g., number of layers/neurons, dropout, and learning rates), and 10% of the simulated data is reserved as validation data used to tune these hyperparameters. The remaining 20% of the simulated data are held out as test data to evaluate the resulting estimates. To estimate the fixed effect β and the functional coefficients $\alpha_{\tau,d}(\cdot, \cdot)$, $d = 1, 2$ under the above three scenarios, while simultaneously demonstrating the efficiency and robustness of our model, We construct two neural network architectures and compare their estimation performance. The first is our proposed FunCQNet model (1), whereas the second is a reduced model denoted by No Interaction in which the structure of α does not incorporate $\mathbf{Z}$.

To evaluate the performance of the estimators, we use the root mean squared errors (RMSEs) for each functional coefficient and the fixed effect parameter β , which are defined as

$$\text{RMSE}(\hat{\alpha}_{\tau,d}) = \sqrt{\int \int [\hat{\alpha}_{\tau,d}(t, \mathbf{Z}) - \alpha_{\tau,d}(t, \mathbf{Z})]^2 dt d\mathbf{Z}},$$

and

$$\text{RMSE}(\hat{\beta}_\tau) = \sqrt{\frac{1}{B} \sum_{b=1}^B (\hat{\beta}_b - \beta_b)^2}.$$

Table 3 summarizes the simulation results of the two estimated functional coefficients $\hat{\alpha}_{\tau,d}(t, \mathbf{Z})$, $d = 1, 2$, and the fixed effect β under various censoring rates when the quantile $\tau = 0.5$. Note that the results using our FunCQNet model outperform the reduced model for all censoring rates under Scenarios 1 and 2. More specifically, it reduces $\text{RMSE}(\hat{\alpha}_{\tau,1})$ by around 92.0% and $\text{RMSE}(\hat{\alpha}_{\tau,2})$ by around 75.0% compared to the method without interaction structure in Scenario 1. Similar results can also be observed in Scenario 2, which demonstrates the superior estimation performance of our model in handling complex data and intricate structures. In Scenario 3, where the true structure of α does not vary with $\mathbf{Z}$, the estimation accuracy of our model remains comparable to that of the simpler model, indicating that our approach achieves robust and reliable estimation without sacrificing performance when the interaction is absent. These improvements remain consistent when the censoring rate increases to 50%. Table 6 reports the prediction loss at quantile level $\tau = 0.5$ under a censoring rate of 30%. It shows that our FunCQNet model also outperforms the reduced model in terms of prediction loss, achieving superior predictive performance. Here, we evaluate the prediction loss by calculating the loss function (5) on the test dataset. In Scenarios 1 and 2, the proposed FunCQNet method achieves approximately 74% and 58% reductions in prediction loss relative to the No Interaction method, while also maintaining comparable performance in Scenario 3.

The RMSEs and prediction loss results for additional quantile levels and censoring rates are provided in Tables 4,5 and Tables 7-17.

Figure 10 illustrates the estimated functional coefficients obtained using three different quantiles and different $\mathbf{Z}$ combinations. As evidenced by these plots, our FunCQNet model demonstrates strong estimation accuracy, with the estimated surfaces or lines aligning closely with their true counterparts. The three-dimensional surfaces of $\alpha_{\tau,1}(t, \mathbf{Z})$ and $\alpha_{\tau,2}(t, \mathbf{Z})$ with $\mathbf{Z}$ fixed as $(1.5, 1.0, 0.0)$ as functions of τ and t are presented in Figure 11.

Table 3. The mean and standard errors (in parentheses) of the root mean squared errors (RMSEs) for the estimated coefficient function $\hat{\alpha}_{\tau,d}(t)$, $d = 1, 2$, and the fixed effect $\hat{\beta}$, respectively, under the quantile $\tau = 0.5$ when varying censoring rates using our proposed FunCQNet model (1) and the reduced model without interaction under three scenarios.

Scenarios	Models		Censoring Rate			
			0%	10%	30%	50%
S1	RMSE($\hat{\alpha}_{\tau,1}$)	FunCQNet	0.028(0.013)	0.029(0.011)	0.040(0.012)	0.057(0.014)
		No Interaction	0.375(0.005)	0.372(0.005)	0.379(0.010)	0.392(0.019)
	RMSE($\hat{\alpha}_{\tau,2}$)	FunCQNet	0.062(0.057)	0.058(0.050)	0.090(0.063)	0.111(0.066)
		No Interaction	0.279(0.017)	0.274(0.013)	0.275(0.017)	0.276(0.019)
	RMSE($\hat{\beta}_{\tau}$)	FunCQNet	0.004(0.004)	0.004(0.003)	0.005(0.005)	0.011(0.008)
		No Interaction	0.070(0.045)	0.068(0.068)	0.064(0.054)	0.060(0.054)
S2	RMSE($\hat{\alpha}_{\tau,1}$)	FunCQNet	0.038(0.015)	0.038(0.017)	0.037(0.012)	0.051(0.017)
		No Interaction	0.370(0.002)	0.371(0.004)	0.371(0.003)	0.381(0.014)
	RMSE($\hat{\alpha}_{\tau,2}$)	FunCQNet	0.094(0.065)	0.107(0.068)	0.090(0.055)	0.122(0.056)
		No Interaction	0.267(0.014)	0.266(0.014)	0.267(0.015)	0.273(0.016)
	RMSE($\hat{\beta}_{\tau}$)	FunCQNet	0.022(0.021)	0.006(0.005)	0.006(0.004)	0.011(0.009)
		No Interaction	0.045(0.045)	0.039(0.036)	0.042(0.041)	0.059(0.056)
S3	RMSE($\hat{\alpha}_{\tau,1}$)	FunCQNet	0.030(0.015)	0.032(0.015)	0.032(0.014)	0.051(0.022)
		No Interaction	0.026(0.009)	0.028(0.013)	0.030(0.018)	0.034(0.015)
	RMSE($\hat{\alpha}_{\tau,2}$)	FunCQNet	0.010(0.091)	0.081(0.070)	0.094(0.082)	0.162(0.097)
		No Interaction	0.057(0.047)	0.104(0.101)	0.094(0.092)	0.072(0.051)
	RMSE($\hat{\beta}_{\tau}$)	FunCQNet	0.010(0.009)	0.009(0.007)	0.015(0.014)	0.014(0.011)
		No Interaction	0.007(0.006)	0.015(0.015)	0.016(0.016)	0.009(0.007)

Table 4. The mean and standard errors (in parentheses) of the root mean squared errors (RMSEs) for the estimated coefficient function $\alpha_d(s, \tau)$, $d = 1, 2$ and the fixed effect β , respectively, under the quantile $\tau = 0.3$ and varying censoring rates using our proposed method and the reduced model under three scenarios.

Models			Censoring Rate			
			0%	10%	30%	50%
S1	RMSE($\hat{\alpha}_1$)	FunCQNet	0.027(0.008)	0.031(0.004)	0.033(0.009)	0.063(0.018)
		No Interaction	0.434(0.013)	0.420(0.011)	0.434(0.010)	0.438(0.019)
	RMSE($\hat{\alpha}_2$)	FunCQNet	0.074(0.066)	0.075(0.068)	0.078(0.066)	0.077(0.064)
		No Interaction	0.311(0.010)	0.312(0.007)	0.309(0.016)	0.311(0.009)
	RMSE($\hat{\beta}$)	FunCQNet	0.051(0.002)	0.050(0.002)	0.051(0.004)	0.056(0.002)
		No Interaction	0.015(0.010)	0.009(0.008)	0.013(0.004)	0.013(0.005)
S2	RMSE($\hat{\alpha}_1$)	FunCQNet	0.030(0.013)	0.035(0.007)	0.041(0.012)	0.055(0.012)
		No Interaction	0.412(0.004)	0.426(0.007)	0.416(0.007)	0.437(0.006)
	RMSE($\hat{\alpha}_2$)	FunCQNet	0.098(0.074)	0.086(0.047)	0.122(0.066)	0.149(0.060)
		No Interaction	0.279(0.014)	0.277(0.010)	0.281(0.016)	0.284(0.015)
	RMSE($\hat{\beta}$)	FunCQNet	0.004(0.004)	0.002(0.002)	0.014(0.009)	0.006(0.003)
		No Interaction	0.060(0.024)	0.044(0.024)	0.054(0.035)	0.048(0.020)
S3	RMSE($\hat{\alpha}_1$)	FunCQNet	0.033(0.014)	0.023(0.012)	0.030(0.010)	0.042(0.008)
		No Interaction	0.031(0.011)	0.027(0.013)	0.021(0.008)	0.026(0.009)
	RMSE($\hat{\alpha}_2$)	FunCQNet	0.096(0.093)	0.085(0.065)	0.100(0.093)	0.151(0.107)
		No Interaction	0.041(0.013)	0.088(0.061)	0.061(0.051)	0.089(0.079)
	RMSE($\hat{\beta}$)	FunCQNet	0.007(0.005)	0.009(0.003)	0.010(0.002)	0.018(0.011)
		No Interaction	0.007(0.004)	0.017(0.012)	0.022(0.013)	0.012(0.004)

Table 5. The mean and standard errors (in parentheses) of the root mean squared errors (RMSEs) for the estimated coefficient function $\alpha_d(s, \tau)$, $d = 1, 2$ and the fixed effect β , respectively, under the quantile $\tau = 0.7$ and varying censoring rates using our proposed method and the reduced model under three scenarios.

Models		Censoring Rate				
		0%	10%	30%	50%	
S1	RMSE($\hat{\alpha}_1$)	FunCQNet	0.023(0.003)	0.041(0.026)	0.056(0.016)	0.060(0.017)
		No Interaction	0.416(0.008)	0.415(0.019)	0.394(0.008)	0.379(0.008)
	RMSE($\hat{\alpha}_2$)	FunCQNet	0.043(0.009)	0.105(0.069)	0.136(0.069)	0.097(0.054)
		No Interaction	0.311(0.013)	0.311(0.012)	0.295(0.014)	0.292(0.022)
	RMSE($\hat{\beta}$)	FunCQNet	0.055(0.001)	0.057(0.008)	0.057(0.005)	0.048(0.007)
		No Interaction	0.023(0.013)	0.017(0.012)	0.103(0.066)	0.203(0.150)
S2	RMSE($\hat{\alpha}_1$)	FunCQNet	0.034(0.015)	0.035(0.008)	0.047(0.014)	0.063(0.017)
		No Interaction	0.417(0.021)	0.411(0.019)	0.388(0.017)	0.430(0.077)
	RMSE($\hat{\alpha}_2$)	FunCQNet	0.086(0.055)	0.069(0.034)	0.146(0.061)	0.176(0.066)
		No Interaction	0.297(0.009)	0.277(0.014)	0.322(0.076)	0.341(0.113)
	RMSE($\hat{\beta}$)	FunCQNet	0.007(0.002)	0.004(0.003)	0.010(0.010)	0.014(0.007)
		No Interaction	0.063(0.059)	0.043(0.015)	0.195(0.183)	0.117(0.105)
S3	RMSE($\hat{\alpha}_1$)	FunCQNet	0.031(0.007)	0.033(0.011)	0.035(0.012)	0.052(0.016)
		No Interaction	0.022(0.008)	0.029(0.012)	0.052(0.029)	0.042(0.015)
	RMSE($\hat{\alpha}_2$)	FunCQNet	0.134(0.074)	0.153(0.104)	0.146(0.094)	0.139(0.063)
		No Interaction	0.091(0.098)	0.145(0.112)	0.139(0.093)	0.108(0.100)
	RMSE($\hat{\beta}$)	FunCQNet	0.007(0.007)	0.008(0.008)	0.018(0.014)	0.014(0.006)
		No Interaction	0.012(0.012)	0.034(0.033)	0.022(0.020)	0.031(0.024)

Table 6. The mean and standard errors (in parentheses) of the prediction loss when the quantile $\tau = 0.5$, and censoring rate= 30% under three scenarios.

Models	Scenarios		
	S1	S2	S3
FunCQNet	0.061(0.003)	0.105(0.002)	0.102(0.004)
No Interaction	0.236(0.003)	0.248(0.003)	0.104(0.004)

Table 7. The mean and standard errors (in parentheses) of the prediction loss when the quantile $\tau = 0.5$, and censoring rate= 0% under three scenarios.

Models	Scenarios		
	S1	S2	S3
FunCQNet	0.063(0.003)	0.113(0.004)	0.113(0.007)
No Interaction	0.268(0.002)	0.277(0.003)	0.111(0.004)

Table 8. The mean and standard errors (in parentheses) of the prediction loss when the quantile $\tau = 0.5$, and censoring rate= 10% under three scenarios.

Models	Scenarios		
	S1	S2	S3
FunCQNet	0.063(0.003)	0.109(0.003)	0.111(0.006)
No Interaction	0.271(0.003)	0.276(0.003)	0.108(0.004)

Table 9. The mean and standard errors (in parentheses) of the prediction loss when the quantile $\tau = 0.5$, and censoring rate= 50% under three scenarios.

Models	Scenarios		
	S1	S2	S3
FunCQNet	0.058(0.002)	0.104(0.003)	0.094(0.004)
No Interaction	0.179(0.002)	0.203(0.004)	0.096(0.004)

Table 10. The mean and standard errors (in parentheses) of the prediction loss when the quantile $\tau = 0.3$, and censoring rate= 0% under three scenarios.

Models	Scenarios		
	S1	S2	S3
FunCQNet	0.044(0.001)	0.080(0.001)	0.081(0.005)
No Interaction	0.170(0.002)	0.187(0.003)	0.079(0.001)

Table 11. The mean and standard errors (in parentheses) of the prediction loss when the quantile $\tau = 0.3$, and censoring rate= 10% under three scenarios.

Models	Scenarios		
	S1	S2	S3
FunCQNet	0.041(0.001)	0.073(0.002)	0.069(0.001)
No Interaction	0.158(0.001)	0.171(0.002)	0.070(0.001)

Table 12. The mean and standard errors (in parentheses) of the prediction loss when the quantile $\tau = 0.3$, and censoring rate= 30% under three scenarios.

Models	Scenarios		
	S1	S2	S3
FunCQNet	0.031(0.001)	0.054(0.003)	0.053(0.003)
No Interaction	0.116(0.002)	0.129(0.002)	0.053(0.001)

Table 13. The mean and standard errors (in parentheses) of the prediction loss when the quantile $\tau = 0.3$, and censoring rate= 50% under three scenarios.

Models	Scenarios		
	S1	S2	S3
FunCQNet	0.022(0.002)	0.038(0.002)	0.038(0.003)
No Interaction	0.069(0.001)	0.082(0.002)	0.037(0.003)

Table 14. The mean and standard errors (in parentheses) of the prediction loss when the quantile $\tau = 0.7$, and censoring rate= 0% under three scenarios.

Models	Scenarios		
	S1	S2	S3
FunCQNet	0.076(0.001)	0.139(0.003)	0.144(0.008)
No Interaction	0.306(0.004)	0.330(0.003)	0.135(0.001)

Table 15. The mean and standard errors (in parentheses) of the prediction loss when the quantile $\tau = 0.7$, and censoring rate= 10% under three scenarios.

Models	Scenarios		
	S1	S2	S3
FunCQNet	0.069(0.005)	0.118(0.004)	0.120(0.007)
No Interaction	0.269(0.001)	0.284(0.007)	0.118(0.005)

Table 16. The mean and standard errors (in parentheses) of the prediction loss when the quantile $\tau = 0.7$, and censoring rate= 30% under three scenarios.

Models	Scenarios		
	S1	S2	S3
FunCQNet	0.053(0.003)	0.087(0.004)	0.079(0.008)
No Interaction	0.180(0.005)	0.192(0.007)	0.082(0.002)

Table 17. The mean and standard errors (in parentheses) of the prediction loss when the quantile $\tau = 0.7$, and censoring rate= 50% under three scenarios.

Models	Scenarios		
	S1	S2	S3
FunCQNet	0.034(0.003)	0.057(0.006)	0.055(0.005)
No Interaction	0.095(0.005)	0.107(0.003)	0.055(0.001)

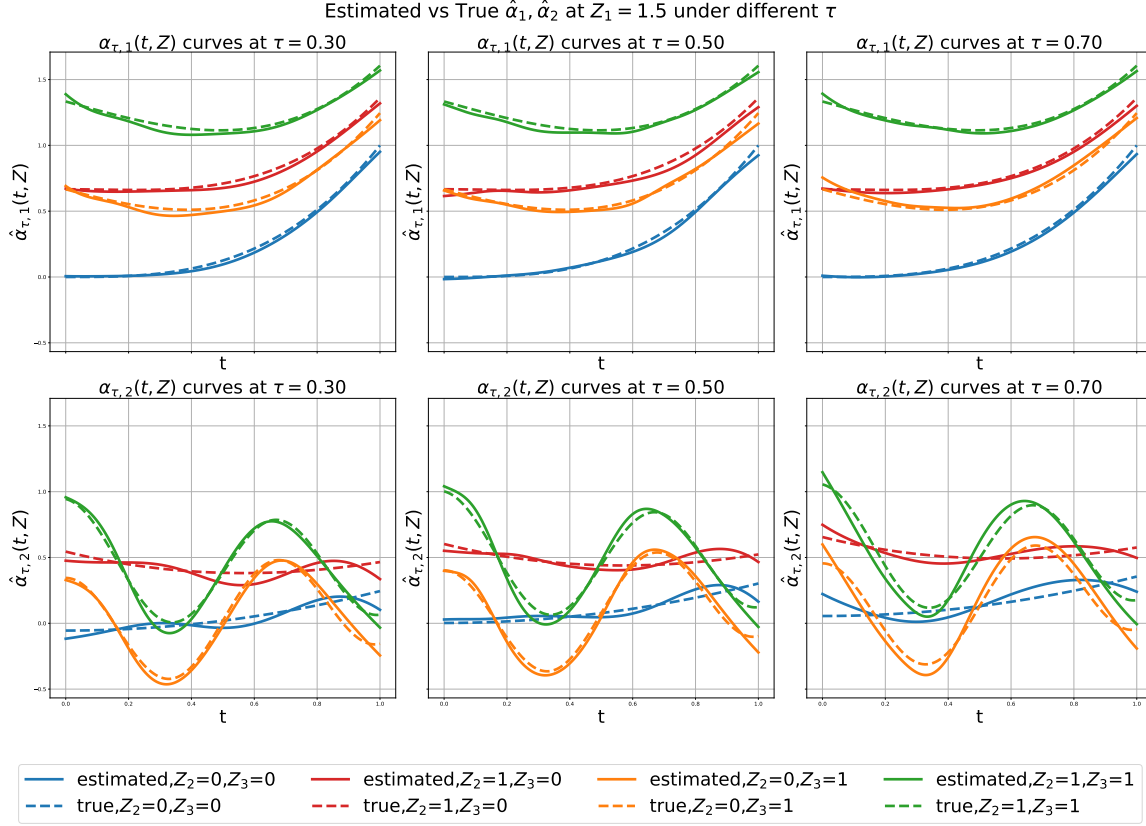


Figure 10. The true functional coefficients $\alpha_{\tau,d}(t, \mathbf{Z})$ and the estimated functional coefficients $\hat{\alpha}_{\tau,d}(t, \mathbf{Z})$, $d = 1, 2$ when the quantile $\tau = 0.3, 0.5, 0.7$, the censoring rate $c = 30\%$, using our proposed FunCQNet model (1) under Scenario 2.

D.2. Simulation II

In this simulation study, we aim to show the performance of our inference. Here, we consider two hypothesis tests: one for the interaction effect of all scalar covariates, and the other for the interaction effect of each scalar covariate:

- **Hypothesis Test for the Interaction Effect of Whole Scalar Covariates:**

$$H_0 : \alpha_\tau(t, \mathbf{Z}) = \alpha_\tau(t), \text{ for any } t \text{ under a given } \tau, \quad (8)$$

We generate the simulated data using

$$\log(T_i) = \beta + \int_0^1 X_{1i}(t) \psi_{11}(t, \mathbf{Z}) dt + \epsilon_i - F_\epsilon^{-1}(\tau), \quad (9)$$

thus $\alpha(t, \mathbf{Z}) = \psi_{11}(t, \mathbf{Z})$, where $\psi_{11}(t, \mathbf{Z}) = 2c/3\{Z_{i1}t^3 + Z_{i2}\exp(-t) + Z_{i3}\cos(t)\} + 1/2t^2$, c varies as $c = 0, 0.1, 0.3, 0.5$, $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma_\epsilon^2)$, and $\sigma_\epsilon^2 = 0.01$, $\beta = 0.5$. F_ϵ is the cumulative distribution function of ϵ . When $c = 0$, Model (9) reduces to a functional censored quantile regression model without interaction. A larger c indicates a greater interaction effect of our covariates on the response.

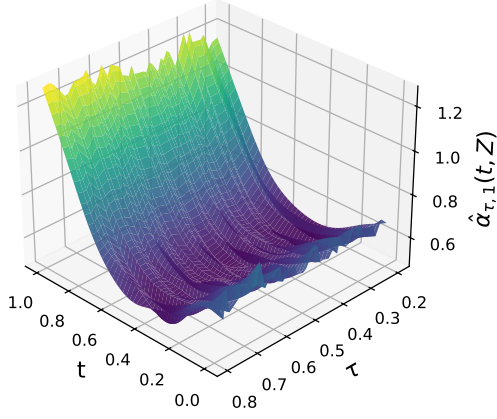
- **Hypothesis Test for the Interaction Effect of Individual Scalar Covariates:**

$$H_{0,j} : \alpha_\tau(t, \mathbf{Z}) = \alpha_\tau(t, \mathbf{Z}_{(-j)}), \text{ for a given } \tau \text{ and any } t, j = 1, \dots, p. \quad (10)$$

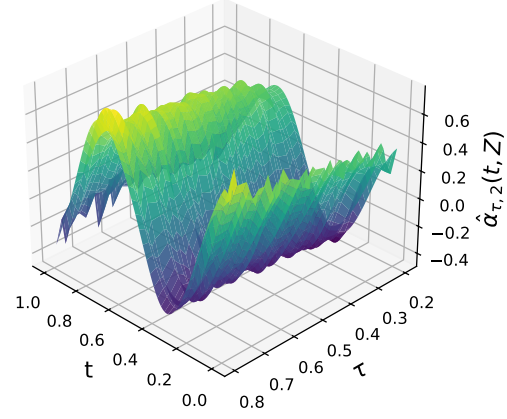
We generate the true data as

$$\log(T_i) = \beta + \int_0^1 X_{1i}(t) \psi_{11}(t, \mathbf{Z}) dt + \epsilon_i - F_\epsilon^{-1}(\tau), \quad (11)$$

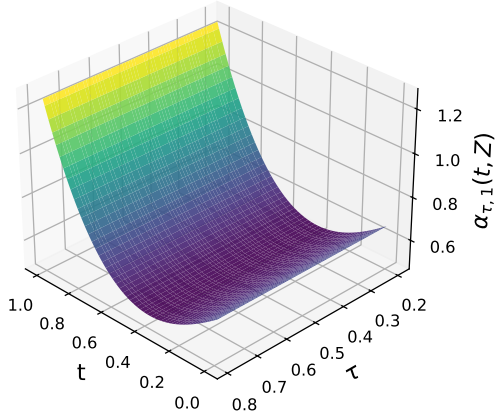
Estimated $\alpha_{\tau,1}(t,Z), Z_1=1.5, Z_2=1, Z_3=0$



Estimated $\alpha_{\tau,2}(t,Z), Z_1=1.5, Z_2=1, Z_3=0$



True $\alpha_{\tau,1}(t,Z), Z_1=1.5, Z_2=1, Z_3=0$



True $\alpha_{\tau,2}(t,Z), Z_1=1.5, Z_2=1, Z_3=0$

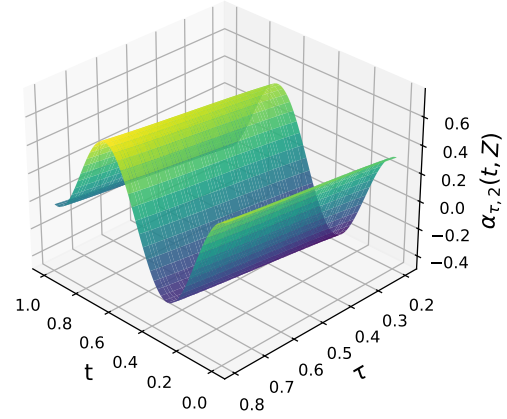


Figure 11. The 3D plots of estimated functional coefficients $\hat{\alpha}_{\tau,d}(t, \mathbf{Z})$ and the true functional coefficients $\alpha_{\tau,d}(t, \mathbf{Z})$, $d = 1, 2$ under different τ , the censoring rate $c = 30\%$, $\mathbf{Z} = (1.5, 1, 0)$, using our proposed FunCQNet model (1) under Scenario 2.

thus $\alpha(t, \mathbf{Z}) = \psi_{11}(t, \mathbf{Z})$, where $\psi_{11}(t, \mathbf{Z}) = 2/3\{cZ_{i1}t^3 + Z_{i2}\exp(-t) + Z_{i3}\cos(t)\}$, c varies as $c = 0, 1, 3, 5$, $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma_\epsilon^2)$, and $\sigma_\epsilon^2 = 0.01$, $\beta = 0.5$. F_ϵ is the cumulative distribution function of ϵ . When $c = 0$, the multi-dimensional coefficient function $\alpha_\tau(t, \mathbf{Z})$ does not depend on the j -th covariate Z_j . A larger c indicates a stronger interaction effect.

The results for these two tests are shown in Figure 12 and Figure 13, respectively. The results indicate that under the null hypothesis, the empirical size remains very close to the nominal level of 0.05, demonstrating good size control of our inference procedure. Moreover, as the value of c increases, the power rises sharply and approaches 1, showing that our method is highly sensitive to departures from the null. For a given c , the power also increases as the censor rate declines, suggesting that the inference procedure becomes more accurate and efficient as more data are available.

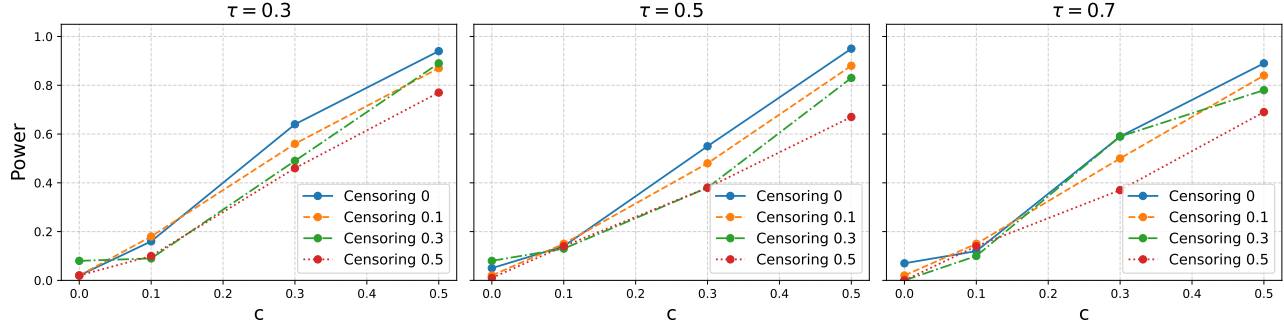


Figure 12. The power of testing for the interaction effect of whole scalar covariates, defined in (8), when the censor rate is 0%, 10%, 30%, 50% and the quantile level $\tau = 0.3, 0.5, 0.7$. A larger c , defined in (9), indicates a stronger interaction effect of the whole scalar covariates on the response.

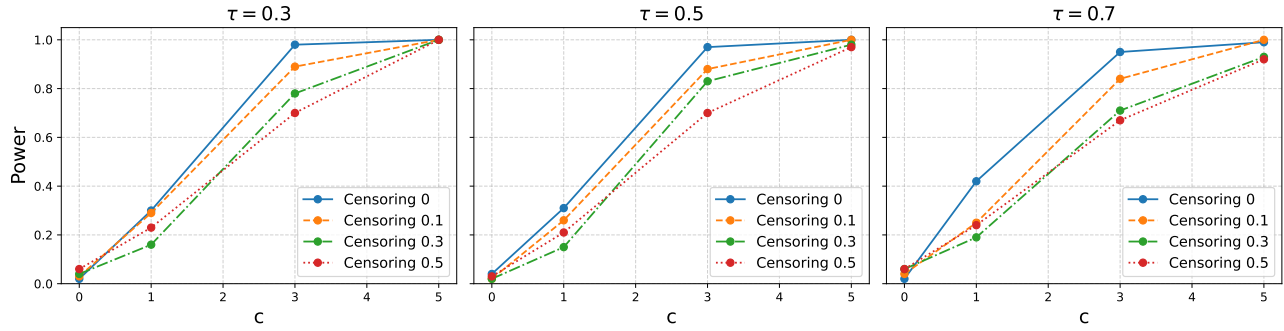


Figure 13. The power of testing for the interaction effect of an individual scalar covariate, defined in (10), when the censor rate is 0%, 10%, 30%, 50% and the quantile level $\tau = 0.3, 0.5, 0.7$. A larger c , defined in (11), indicates a stronger interaction effect of an individual scalar covariate on the response.